

Figure S.3.25: Figure for Exercise 8e of Sec. 3.4.

We need to integrate this function over the set where $x > y$. That region can be written as $\{(x, y) : 4 < x < 200, 1 < y < \min\{x, 150\}\}$. The reason for the complicated upper limit on y is that we require both $y < x$ and $y < 150$.

$$\begin{aligned}
\int_4^{200} \int_1^{\min\{x,150\}} \frac{1}{29204} dy dx &= \int_4^{200} \frac{\min\{x-1, 149\}}{29204} dx \\
&= \int_4^{150} \frac{x-1}{29204} dx + \int_{150}^{200} \frac{149}{29204} dx \\
&= \left. \frac{(x-1)^2}{2 \times 29204} \right|_{x=4}^{150} + \frac{50 \times 149}{29204} \\
&= \frac{149^2 - 3^2}{58408} + \frac{7450}{29204} = 0.63505.
\end{aligned}$$

10. (a) The sum of $f(x, y)$ over all x for each fixed y is

$$\exp(-3y) \sum_{x=0}^{\infty} \frac{(2y)^2}{x!} = \exp(-3y) \exp(2y) = \exp(-y),$$

where the first equality follows from the power series expansion of $\exp(2y)$. The integral of the resulting sum is easily calculated to be 1.

(b) We can compute $\Pr(X = 0)$ by integrating $f(0, y)$ over all y:

$$\Pr(X = 0) = \int_0^{\infty} \frac{(2y)^0}{0!} \exp(-3y) dy = \frac{1}{3}.$$

11. Let $f(x, y)$ stand for the joint p.f. in Table 3.3 in the text for $x = 0, 1$ and $y = 1, 2, 3, 4$.

(a) We are asked for the probability for the set $\{Y \in \{2, 3\}\} \cap \{X = 1\}$, which is $f(1, 2) + f(1, 3) = 0.166 + 0.107 = 0.273$.

(b) This time, we want $\Pr(X = 0) = f(0, 1) + f(0, 2) + f(0, 3) + f(0, 4) = 0.513$.

3.5 Marginal Distributions

Commentary

Students can get confused when solving problems like Exercises 7 and 8 in this section. They notice that the functional form of $f(x, y)$ factors into $g_1(x)g_2(y)$ for those (x, y) pairs such that $f(x, y) > 0$, but they don't understand that the factorization needs to hold even for those (x, y) pairs such that $f(x, y) = 0$. When the

two marginal p.d.f.'s are both strictly positive on intervals, then the set of (x, y) pairs where $f_1(x)f_2(y) > 0$ must be a rectangle (with sides parallel to the coordinate axes), even if the rectangle is infinite in one or more directions. Hence, it is a *necessary* condition for independence that the set of (x, y) pairs such that $f(x, y) > 0$ be a rectangle with sides parallel to the coordinate axes. Of course, it is also necessary that $f(x, y) = g_1(x)g_2(y)$ for those (x, y) such that $f(x, y) > 0$. The two necessary conditions together are sufficient to insure independence, but neither is sufficient alone. See the solution to Exercise 8 below for an illustration of how to illustrate that point.

Solutions to Exercises

1. The joint p.d.f. is constant over a rectangle with sides parallel to the coordinate axes. So, for each x, the integral over y will equal the constant times the length of the interval of y values, namely $d - c$. Similarly, for each y, the integral over x will equal the constant times the length of the interval of x values, namely $b - a$. Of course the constant k must equal one over the area of the rectangle. So $k = 1/[(b - a)(d - c)]$. So the marginal p.d.f.'s of X and Y are

$$f_1(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a \le x \le b, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_2(y) = \begin{cases} \dfrac{1}{d-c} & \text{for } c \le y \le d, \\ 0 & \text{otherwise.} \end{cases}$$

2. (a) For $x = 0, 1, 2$, we have

$$f_1(x) = \sum_{y=0}^{3} f(x, y) = \frac{1}{30}(4x + 6) = \frac{1}{15}(2x + 3).$$

Similarly, for $y = 0, 1, 2, 3$, we have

$$f_2(y) = \sum_{x=0}^{2} f(x, y) = \frac{1}{30}(3 + 3y) = \frac{1}{10}(1 + y).$$

(b) X and Y are not independent because it is not true that $f(x, y) = f_1(x)f_2(y)$ for all possible values of x and y.

3. (a) For $0 \le x \le 2$, we have

$$f_1(x) = \int_0^1 f(x, y)\, dy = \frac{1}{2}.$$

Also, $f_1(x) = 0$ for x outside the interval $0 \le x \le 2$. Similarly, for $0 \le y \le 1$,

$$f_2(y) = \int_0^2 f(x, y)\, dx = 3y^2.$$

Also, $f_2(y) = 0$ for y outside the interval $0 \le y \le 1$.

(b) X and Y are independent because $f(x, y) = f_1(x)\ f_2(y)$ for $-\infty < x < \infty$ and $-\infty < y < \infty$.

(c) We have

$$\Pr\left(X < 1 \text{ and } Y \ge \frac{1}{2}\right) = \int_0^1 \int_{1/2}^1 f(x, y)\, dx\, dy$$

$$= \int_0^1 \int_{1/2}^1 f_1(x) f_2(y)\, dx\, dy$$
$$= \int_0^1 f_1(x)\, dx \int_{1/2}^1 f_2(y)\, dy = \Pr(X < 1) \Pr\left(Y > \frac{1}{2}\right).$$

Therefore, by the definition of the independence of two events (Definition 2.2.1), the two given events are independent.

We can also reach this answer, without carrying out the above calculation, by reasoning as follows: Since the random variables X and Y are independent, and since the occurrence or nonoccurence of the event $\{X < 1\}$ depends on the value of X only while the occurrence or nonoccurence of the event $\{Y \geq 1/2\}$ depends on the value of Y only, it follows that these two events must be independent.

4. (a) The region where $f(x, y)$ is non-zero is the shaded region in Fig. S.3.26. It can be seen that the

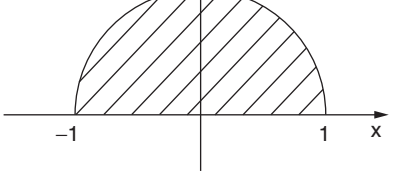


Figure S.3.26: Figure for Exercise 4a of Sec. 3.5.

possible values of X are confined to the interval $-1 \leq X \leq 1$. Hence, $f_1(x) = 0$ for values of x outside this interval. For $-1 \leq x \leq 1$, we have

$$f_1(x) = \int_0^{1-x^2} f(x, y)\, dy = \frac{15}{4} x^2 (1 - x^2).$$

Similarly, it can be seen from the sketch that the possible values of Y are confined to the interval $0 \leq Y \leq 1$. Hence, $f_2(y) = 0$ for values of y outside this interval. For $0 \leq y \leq 1$, we have

$$f_2(y) = \int_{-(1-y)^{1/2}}^{(1-y)^{1/2}} f(x, y)\, dx = \frac{5}{2}(1 - y)^{3/2}.$$

(b) X and Y are not independent because $f(x, y) \neq f_1(x) f_2(y)$.

5. (a) Since X and Y are independent,

$$f(x, y) = \Pr(X = x \text{ and } Y = y) = \Pr(X = x) \Pr(Y = y) = p_x p_y.$$

(b) $\Pr(X = Y) = \sum_{i=0}^{3} f(i, i) = \sum_{i=0}^{3} p_i^2 = 0.3.$

(c) $\Pr(X > Y) = f(1, 0) + f(2, 0) + f(3, 0) + f(2, 1) + f(3, 1) + f(3, 2) = 0.35.$

6. (a) Since X and Y are independent

$$f(x, y) = f_1(x)\, f_2(y) = g(x) g(y) = \begin{cases} \frac{9}{64} x^2 y^2 & \text{for } 0 \leq x \leq 2,\ 0 \leq y \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$

(b) Since X and Y have a continuous joint distribution, $\Pr(X = Y) = 0$.

(c) Since X and Y are independent random variables with the same probability distribution, it must be true that $\Pr(X > Y) = \Pr(Y > X)$. Since $\Pr(X = Y) = 0$, it therefore follows that $\Pr(X > Y) = 1/2$.

(d) $\Pr(X + Y \le 1) = \Pr(\text{shaded region in sketch})$

$$= \int_0^1 \int_0^{1-y} f(x, y)\, dx\, dy = \frac{1}{1280}.$$

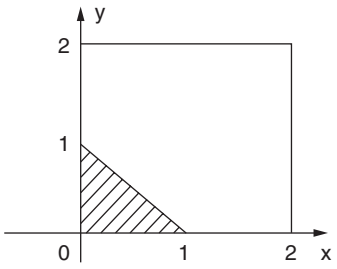


Figure S.3.27: Figure for Exercise 6d of Sec. 3.5.

7. Since $f(x, y) = 0$ outside a rectangle and $f(x, y)$ can be factored as in Eq. (3.5.7) inside the rectangle (use $h_1(x) = 2x$ and $h_2(y) = \exp(-y)$), it follows that X and Y are independent.

8. Although $f(x, y)$ can be factored as in Eq. (3.5.7) inside the triangle where $f(x, y) > 0$, the fact that $f(x, y) > 0$ inside a triangle, rather than a rectangle, implies that X and Y cannot be independent. (Note that $y \ge 0$ should have appeared as part of the condition for $f(x, y) > 0$ in the statement of the exercise.) For example, to factor $f(x, y)$ as in Eq. (3.5.7) we write $f(x, y) = g_1(x)g_2(y)$. Since $f(1/3, 1/4) = 2$ and $f(1/6, 3/4) = 3$, it must be that $g_1(1/3) > 0$ and $g_2(3/4) > 0$. However, since $f(1/3, 3/4) = 0$, it must be that either $g_1(1/3) = 0$ or $g_2(3/4) = 0$. These facts contradict each other, hence f cannot have a factorization as in (3.5.7).

9. (a) Since $f(x, y)$ is constant over the rectangle S and the area of S is 6 units, it follows that $f(x, y) = 1/6$ inside S and $f(x, y) = 0$ outside S. Next, for $0 \le x \le 2$,

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y)\, dy = \int_1^4 \frac{1}{6} dy = \frac{1}{2}.$$

Also, $f_1(x) = 0$ otherwise. Similarly, for $1 \le y \le 4$,

$$f_2(y) = \int_0^2 \frac{1}{6}\, dx = \frac{1}{3}.$$

Also, $f_2(y) = 0$ otherwise. Thus, the marginal distribution of both X and Y are uniform distributions.

(b) Since $f(x, y) = f_1(x)\, f_2(y)$ for all values of x and y, it follows that X and Y are independent.

10. (a) $f(x, y)$ is constant over the circle S in Fig. S.3.28. The area of S is π units, and it follows that $f(x, y) = 1/\pi$ inside S and $f(x, y) = 0$ outside S. Next, the possible values of x range from -1 to 1. For any value of x in this interval, $f(x, y) > 0$ only for values of y between $-(1 - x^2)^{1/2}$ and $(1 - x^2)^{1/2}$. Hence, for $-1 \le x \le 1$,

$$f_1(x) = \int_{-(1-x^2)^{1/2}}^{(1-x^2)^{1/2}} \frac{1}{\pi} dy = \frac{2}{\pi}(1 - x^2)^{1/2}.$$

Also, $f_1(x) = 0$ otherwise. By symmetry, the random variable Y will have the same marginal p.d.f. as X.

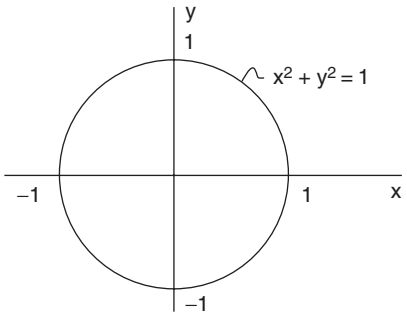


Figure S.3.28: Figure for Exercise 10 of Sec. 3.5.

(b) Since $f(x, y) \neq f_1(x)f_2(y)$, X and Y are not independent.
The conclusions found in this exercise in which X and Y have a uniform distribution over a circle should be contrasted with the conclusions found in Exercise 9, in which X and Y had a uniform distribution over a rectangle with sides parallel to the axes.

11. Let X and Y denote the arrival times of the two persons, measured in terms of the number of minutes after 5 P.M. Then X and Y each have the uniform distribution on the interval (0, 60) and they are independent. Therefore, the joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} \dfrac{1}{3600} & \text{for } 0 < x < 60,\ 0 < y < 60, \\ 0 & \text{otherwise.} \end{cases}$$

We must calculate $\Pr(|X - Y| < 10)$, which is equal to the probability that the point (X, Y) lies in the shaded region in Fig. S.3.29. Since the joint p.d.f. of X and Y is constant over the entire square, this

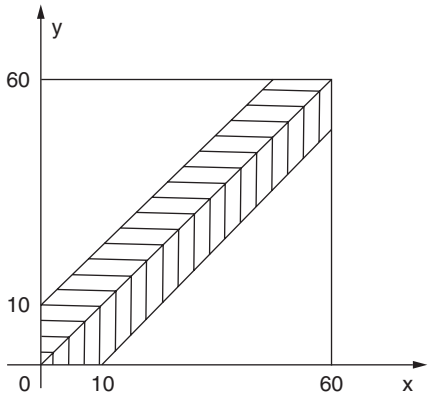


Figure S.3.29: Figure for Exercise 11 of Sec. 3.5.

probability is equal to (area of shaded region)/3600. The area of the shaded region is 1100. Therefore, the required probability is $1100/3600 = 11/36$.

12. Let the rectangular region be $R = \{(x, y) : x_0 < x < x_1, y_0 < y < y_1\}$ with x_0 and/or y_0 possibly $-\infty$ and x_1 and/or y_1 possibly ∞. For the "if" direction, assume that $f(x, y) = h_1(x)h_2(y)$ for all (x, y) that satisfy $f(x, y) > 0$. Then define

$$h_1^*(x) = \begin{cases} h_1(x) & \text{if } x_0 < x < x_1, \\ 0 & \text{otherwise.} \end{cases}$$

$$h_2^*(y) = \begin{cases} h_2(y) & \text{if } y_0 < y < y_1, \\ 0 & \text{otherwise.} \end{cases}$$

Then $h_1^*(x)h_2^*(y) = h_1(x)h_2(y) = f(x,y)$ for all $(x,y) \in R$ and $h_1^*(x)h_2^*(y) = 0 = f(x,y)$ for all $(x,y) \notin R$. Hence $f(x,y) = h_1^*(x)h_2^*(y)$ for all (x,y), and X and Y are independent.

For the "only if" direction, assume that X and Y are independent. According to Theorem 3.5.5, $f(x,y) = h_1(x)h_2(y)$ for all (x,y). Then $f(x,y) = h_1(x)h_2(y)$ for all $(x,y) \in R$.

13. Since $f(x,y) = f(y,x)$ for all (x,y), it follows that the marginal p.d.f.'s will be the same. Each of those marginals will equal the integral of $f(x,y)$ over the other variable. For example, to find $f_1(x)$, note that for each x, the values of y such that $f(x,y) > 0$ form the interval $[-\sqrt{1-x^2}, \sqrt{1-x^2}]$. Then, for $-1 \le x \le 1$,

$$\begin{aligned} f_1(x) &= \int f(x,y)dy \\ &= \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} kx^2y^2 dy \\ &= kx^2 \left.\frac{y^3}{3}\right|_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \\ &= 2kx^2(1-x^2)^{3/2}/3. \end{aligned}$$

14. The set in Fig. 3.12 is not rectangular, so X and Y are *not* independent.

15. (a) Figure S.3.30 shows the region where $f(x,y) > 0$ as the union of two shaded rectangles. Although the region is not a rectangle, it is a *product set*. That is, it has the form $\{(x,y) : x \in A, y \in B\}$ for two sets A and B of real numbers.

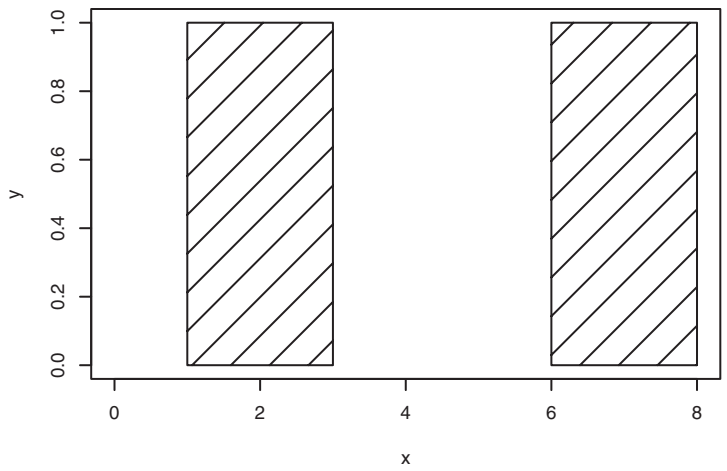


Figure S.3.30: Region of positive p.d.f. for Exercise 15a of Sec. 3.5.

(b) The marginal p.d.f. of X is

$$f_1(x) = \int_0^1 f(x,y)dy = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3, \\ \frac{1}{6} & \text{if } 6 < x < 8. \end{cases}$$

The marginal p.d.f. of Y is

$$f_2(y) = \int_1^3 \frac{1}{3}dx + \int_6^8 \frac{1}{6}dx = 1,$$

for $0 < y < 1$. The distribution of Y is the uniform distribution on the interval $[0, 1]$.

(c) The product of the two marginal p.d.f.'s is

$$f_1(x)f_2(y) = \begin{cases} \frac{1}{3} & \text{if } 1 < x < 3 \text{ and } 0 < y < 1, \\ \frac{1}{6} & \text{if } 6 < x < 8 \text{ and } 0 < y < 1, \\ 0 & \text{otherwise,} \end{cases}$$

which is the same as $f(x, y)$, hence the two random variables are independent. Although the region where $f(x, y) > 0$ is not a rectangle, it is a product set as we saw in part (a). Although it is sufficient in Theorem 3.5.6 for the region where $f(x, y) > 0$ to be a rectangle, it is necessary that the region be a product set. Technically, it is necessary that there is a version of the p.d.f. that is strictly positive on a product set. For continuous joint distributions, one can set the p.d.f. to arbitrary values on arbitrary one-dimensional curves without changing it's being a joint p.d.f.

3.6 Conditional Distributions

Commentary

When introducing conditional distributions given continuous random variables, it is important to stress that we are not conditioning on a set of 0 probability, even if the popular notation makes it appear that way. The note on page 146 can be helpful for students who understand two-variable calculus. Also, Exercise 25 in Sec. 3.11 can provide additional motivation for the idea that the conditional distribution of X given $Y = y$ is really a surrogate for the conditional distribution of X given that Y is close to y, but we don't wish to say precisely how close. Exercise 26 in Sec. 3.11 (the Borel paradox) brings home the point that conditional distributions really are not conditional on the probability 0 events such as $\{Y = y\}$.

Also, it is useful to stress that conditional distributions behave just like distributions. In particular, conditional probabilities can be calculated from conditional p.f.'s and conditional p.d.f.'s in the same way that probabilities are calculated from p.f.'s and p.d.f.'s. Also, be sure to advertise that all future concepts and theorems will have conditional versions that behave just like the marginal versions.

Solutions to Exercises

1. We begin by finding the marginal p.d.f. of Y. The set of x values for which $f(x, y) > 0$ is the interval $[-(1-y^2)^{1/2}, (1-y^2)^{1/2}]$. So, the marginal p.d.f. of Y is, for $-1 \le y \le 1$,

$$f_2(y) = \int_{-(1-y^2)^{1/2}}^{(1-y^2)^{1/2}} kx^2y^2dx = \left.\frac{ky^2}{3}x^3\right|_{x=-(1-y^2)^{1/2}}^{(1-y^2)^{1/2}} = \frac{2k}{3}y^2(1-y^2)^{3/2},$$

and 0 otherwise. The conditional p.d.f. of X given $Y = y$ is the ratio of the joint p.d.f. to the marginal p.d.f. just found.

$$g_1(x|y) = \begin{cases} \dfrac{3x^2}{2(1-y^2)^{3/2}} & \text{for } -(1-y^2)^{1/2} \le x \le (1-y^2)^{1/2}, \\ 0 & \text{otherwise.} \end{cases}$$